LLS - Low Light Signal Measurements

Signature Sheet

Student's Name _____ Partner's Name _____

Suggested reading to start with (see the end of the manual for more):

1. 111B LLS Manual
2. About Lock-In Amplifiers: Application Note # 3" **About Lock-ins**
3. R. Ramirez, "**The FFT: Fundamentals and Concepts**"; Prentice-Hall, 1985, pp. 3-16 & 115-123.
4. SRS **SR830 Lock-In Amplifier Operating Manual**, "SR830 Basics" chapter — in particular the TIME CONSTANT, SLOPE, and ENBW discussion on p. 3-21.
5. 111B LLS Experiment Videos **Low Light Signal Measurements** and the **Light Sources and Detectors**. Note: In order to view the private Youtube videos hosted by the university, you must be signed into your berkeley.edu Google account.

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. What is the purpose of having the SR760 FFT Spectrum Analyzer? What mathematical operation does it perform on an input signal? What determines its frequency *resolution*, and what determines its frequency *range* (the maximum SPAN)? Roughly how long do you need to record the signal to resolve two peaks 100 mHz apart? (After question 4, come back and make the same rough estimate for the lock-in: with a 100 mHz detection bandwidth, how long does the measurement take? See the TIME CONSTANT, SLOPE, and wait time discussion in the SR830 manual, p. 3-21.)
2. Why don't we just put a filter in front of our photodiode which only passes light at the frequency given off by the LED (e.g. 635 nm)?
3. Explain how chopping the steady-state signal from the LED helps you to recover the signal from the LED.
4. Explain the working principle of lock-in amplifiers. You can first assume that the input signal is an arbitrary signal expressed by a Fourier Series, and assume your reference signal is a unit sine wave at targeted frequency. Show how the target frequency component of the input signal is extracted. Consider how the phase difference between the input and reference signal and lock-in settings such as the SLOPE and TIME CONSTANT affect the result. (Hint: Think about the relation between X, Y, and R).
5. Here is a scheme you will meet in real research: a *sideband measurement*. A mechanical resonator driven at $\Omega/2\pi = 100$ kHz is read out electrically at $\omega/2\pi = 10$ kHz, so the voltage at your detector is the *product*

$$V_{sig}(t) \;=\; A\cos(\Omega t)\cos(\omega t).$$

(Not in this lab, but common in practice: think of a power-modulated laser chopped by a mechanical chopper wheel, or an AFM tip tapping on a sample under an AC voltage bias — in both cases the two modulations multiply.) At which frequencies does the signal actually appear? What would a single lock-in referenced to 100 kHz (or 10 kHz) read? You have reference outputs at 100 kHz and at 10

kHz, but none at 110 kHz — show how you can still measure A with *two* lock-ins in series. (Hint: give the first lock-in a *short* TIME CONSTANT, so that its output still oscillates — at which frequency? Should you feed its X or its R output into the second lock-in?)

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. Before beginning Experiment X, you should have produced a plot of the Johnson noise signal using the measurements obtained in Experiment IX.

Staff Signature _____ Date _____

Completed before proceeding with Experiment X? (Circle one) Yes / No

Checkpoint Signatures

1. Fourier Components

   Staff Signature _____

2. Slope/Oct and Time Constant

   Staff Signature _____

3. 1/f Noise

Staff Signature _____

4. Capacitive Noise

Staff Signature _____

5. Johnson-Noise Measurement

Staff Signature _____